

Hypothesis Stress-Test & Investment Decision

Report v3.0.6.8

Powered by Vraimony

Protocol: ImmuneErrorRadar falsification protocol v3.0.6.9.7 | method=Vraimony methodology
| posture=standard_followed_not_copied

Falsification receipt: id=ierfp-0a55bc869b4572e87aed1220 |
contentSha256=0a55bc869b4572e87aed1220 | signedByVraimony=false

Documentation standard: ERF-inspired evidence receipt discipline: hash-first, portable,
replayable, claim-bounded.

Citation line: This report follows the ImmuneErrorRadar falsification protocol, powered by
Vraimony, based on the Vraimony evidence-receipt methodology.

Vraimony boundary: This receipt documents a falsification workflow and report payload
hash; it is not medical advice, not clinical validation, not wet-lab evidence, and not
proof of biology.

Report mode: verification-only | evidence-readiness receipt | investment-triage verdict |
no discovery claim | no clinical advice

FINAL INVESTMENT DECISION v3.0.6.7: DO_NOT_INVEST_CLAIM_KILLED | claim=claim killed as
written for current data |

killedClaim=bulk_selected_matrix_can_test_NECTIN2_TIGIT_ligand_receptor_checkpoint_function
| criteria=kill:5/invest:0 | next=Do not invest in bulk NECTIN2/TIGIT claim. Only a
single-cell/functional checkpoint dataset can reopen it. | probabilityReported=false

PREPRINT READINESS v3.0.6.8: PREPRINT_READY_CLAIM_KILLED_AS_WRITTEN | title=Hypothesis
Stress-Test & Investment Decision Report v3.0.6.8 | killedAsWritten=true |
residualDataFollowUp=false | wetLabAllowed=false | probabilityReported=false

Publication data fuel v3.0.6.9: route=NECTIN2_TIGIT_NOT_MEASURABLE |
state=NECTIN2_TIGIT_SINGLE_CELL_FUNCTIONAL_REOPEN_GATES_INJECTED_BULK_CLAIM_KILLED |
syntheticData=false | fakeReplication=false | defense=NECTIN2/TIGIT bulk claim is killed
because the relevant checkpoint biology is ligand-receptor and single-cell/functional, not
bulk selected-matrix Cox.

Pediatric cross-tumor scope v3.0.6.9.6: state=NEUROBLASTOMA_ONLY_SINGLE_COHORT_SCOPE |
crossTumorPromotionAllowed=false | wetLabPromotionAllowed=false

Required external pediatric tumors: medulloblastoma,Wilms tumor / nephroblastoma,Ewing
sarcoma

Investment triage v3.0.6.6: verdict=KILL_CURRENT_BULK_CHECKPOINT_CLAIM |
killedClaim=bulk_selected_matrix_can_test_NECTIN2_TIGIT_ligand_receptor_checkpoint_function
| basis=kill criteria dominate and no defensible next investment exists for the supplied
claim; kill=5, invest=0 | next=Do not invest in bulk NECTIN2/TIGIT claim. Only a
single-cell/functional checkpoint dataset can reopen it. | probabilityReported=false

Question understood: true

Question: In pediatric neuroblastoma GSE85047, high NECTIN2 (PVRL2) expression on tumor
cells engages TIGIT on infiltrating T and NK cells, suppressing immune surveillance.

Tumors with high NECTIN2 expression show worse progression-free survival independently of
MYCN amplification, identifying the NECTIN2-TIGIT axis as a functional immune checkpoint
in this cohort

Question understanding v3.0.3.97: NECTIN2_TIGIT_IMMUNE_CHECKPOINT | confidence=0.89

Mechanistic text understanding v3.0.4.3: NECTIN2_TIGIT_NEUROBLASTOMA_CHECKPOINT_TEXT |
confidence=0.96

Text source context: Princess Maxima/Cancer Cell NECTIN2-TIGIT style text |
prior=SUPPORTED_NEUROBLASTOMA_CHECKPOINT_AXIS
Text genes understood: NECTIN2,TIGIT,CD226,CD96,NKG7,GZMB
Mechanism claim parsed: NECTIN2/PVRL2 on tumor cells may engage TIGIT on T/NK cells and suppress cytotoxic surveillance.
Local proxy tests:
NECTIN2_TIGIT_BULK_PROXY:BULK_PROXY_ONLY_RECEPTOR_ENGAGEMENT_NOT_TESTABLE
Not directly testable here: ligand-receptor engagement,TIGIT receptor function on single T/NK cells,exhaustion phenotype validation,checkpoint blockade response
Why this route: NECTIN2,PVRL2,TIGIT,checkpoint
Cancer genes understood: MYCN,NECTIN2,TIGIT,CD226,CD96,NKG7,GZMB
Dataset/outcome resolved: GSE85047 | PFS | covariates=MYCN
Literature axis basis: NECTIN2/PVRL2-TIGIT text is treated as a ligand-receptor checkpoint prior; bulk expression can only test proxy direction.
Understanding caveats: ligand_receptor_engagement_requires_single_cell_or_functional_assay
Hypothesis tested: NECTIN2_TIGIT_CHECKPOINT_PFS_MYCN_GSE85047
Primary axis: NECTIN2_TIGIT_CHECKPOINT_PROXY_NOT_EXECUTABLE |
route=PRIMARY_AXIS_NECTIN2_TIGIT_CHECKPOINT_TEXT_PROXY
Genes understood: MYCN,NECTIN2,TIGIT,CD226,CD96,NKG7,GZMB
Dataset/model: GSE85047 | outcome=PFS | model=NECTIN2/TIGIT checkpoint text proxy – not executable until NECTIN2/TIGIT/CD226/CD96 probes are mapped
Initial result: NO_SELECTED_MATRIX_RESULT_AVAILABLE | priority=REVIEW_ONLY
Triage signal: REVIEW_ONLY_DIRECTIONAL_SCREEN | level=REVIEW_ONLY_SIGNAL_NOT_AVAILABLE |
verdict=NO_REVIEW_ONLY_SIGNAL_AVAILABLE_OFFICIAL_CLAIM_BLOCKED
Signal promotion: rawAllowed=false | forKilledClaimAllowed=false |
residualDataFollowUpAllowed=false | nextComputationalRoundCandidate=false |
wetLabCandidate=false | officialClaimAllowed=false
Directional decision v3.0.4.1: NOT_TESTABLE_WITH_CURRENT_DATA |
local=NO_LOCAL_DECISION_DATA_ASSET_INSUFFICIENT | rank=12
Best direction scan: APM_COMPOSITE_PFS_MYCN_GSE85047 | STRONG_EXPLORATORY_DIRECTION |
score=12.3647
Directional ranking table: #1 APM composite:STRONG_EXPLORATORY_DIRECTION | #2 B2M/TAP1/NK:STRONG_EXPLORATORY_DIRECTION | #3 CK2/APM
fallback:APM_FALLBACK_SIGNAL_MECHANISM_NOT_DIRECTLY_TESTED | #4 HDAC/APM
fallback:APM_FALLBACK_SIGNAL_MECHANISM_NOT_DIRECTLY_TESTED | #5 NK
effector:DIRECTIONAL_LEAD_CONFOUNDER_SENSITIVE | #6
γδ/NKG2D:GAMMA_DELTA_SURROGATE_DIRECTION_TRGC_TRDV_FULL_MAPPING_REQUIRED | #7
CD276/B7-H3:SUBGROUP_ONLY_DIRECTION
Lead label: NECTIN2_TIGIT_NOT_TESTABLE_PROBE_AND_SINGLE_CELL_CONTEXT_REQUIRED
Directional why: Selected-matrix rows/events are insufficient or route did not execute.
Directional next action: Fix routing, selected probes, or source-certified tiles before judging direction.
Proxy attribution v3.0.4.4: NOT_TESTABLE_PROBE_MISSING |
signalSource=DIRECT_OR_ROUTE_SPECIFIC_SELECTED_MATRIX_PROXY | mechanismGeneTested=false |
label=NECTIN2_TIGIT_NOT_TESTABLE_PROBE_AND_SINGLE_CELL_CONTEXT_REQUIRED
Probe lookup v3.0.5.0: resolvedVector=8 | resolvedButVectorMissing=20 | probeUnknown=0 |
absentOrReporterUnavailable=2 |
staticManifestLocalResolve=TARGETED_STATIC_MANIFEST_ATTACHED | runtimeNetwork=false |
legacyAutoFetchFlag=true
Probe vector binding v3.0.5.1: columnIndexPresent=true | resolvedProbelds=28 |

boundVectors=8 | resolvedButColumnMissing=20 | aliasBoundVectors=0 |
nextBlocker=PARTIAL_BINDING_REMAINING_GENES_NEED_COLUMN_EXTRACTION
Evidence fuel v3.0.6.0: NECTIN2_TIGIT_NOT_MEASURABLE |
ROUTE_UNDERSTOOD_BUT_PROBES_AND_SINGLE_CELL_CONTEXT_MISSING |
expected=NOT_MEASURABLE_PROBE_AND_CONTEXT_BLOCKED |
boundary=DATA_FUEL_IS_TEST_RECIPE_AND_SELECTED_RUNTIME_COVERAGE_NOT_TIER2_PROOF
Executed local directions v3.0.4.4: #1 APM
composite:STRONG_EXPLORATORY_DIRECTION[APM_COMPOSITE_DIRECT_PROXY] | #2
B2M/TAP1/NK:STRONG_EXPLORATORY_DIRECTION[B2M_TAP1_NK_DIRECT_PROXY] | #5 NK
effector:DIRECTIONAL_LEAD_CONFOUNDER_SENSITIVE[DIRECT_OR_ROUTE_SPECIFIC_SELECTED_MATRIX_PROXY]
| #6
yδ/NKG2D:GAMMA_DELTA_SURROGATE_DIRECTION_TRGC_TRDV_FULL_MAPPING_REQUIRED[GAMMA_DELTA_S
| #7 CD276/B7-H3:SUBGROUP_ONLY_DIRECTION[DIRECT_OR_ROUTE_SPECIFIC_SELECTED_MATRIX_PROXY] |
#8
ODC1/polyamine:WEAK_EXPLORATORY_DIRECTION[DIRECT_OR_ROUTE_SPECIFIC_SELECTED_MATRIX_PROXY]
Mechanistic hypotheses needing assets v3.0.4.4: CK2/APM
fallback:APM_FALLBACK_STRONG_SIGNAL_CK2_MECHANISM_NOT_DIRECTLY_TESTED | HDAC/APM
fallback:APM_FALLBACK_STRONG_SIGNAL_EPIGENETIC_MECHANISM_NOT_DIRECTLY_TESTED |
yδ/NKG2D:GAMMA_DELTA_SURROGATE_DIRECTION_TRGC_TRDV_FULL_MAPPING_REQUIRED |
NECTIN2/TIGIT:NECTIN2_TIGIT_NOT_TESTABLE_PROBE_AND_SINGLE_CELL_CONTEXT_REQUIRED
N=0 | events=0 | highEventRate=0.0 | lowEventRate=0.0 | delta=0.0
Cox first-pass: not executed
MYCN-adjusted Cox first-pass: not executed
MYCN-stratified screen: not executed
Component separation: not applicable
Tier-1 package v3.0.4.2: TIER1_FULL_GPL5175_LOCAL_COX_TARGET |
current=TIER0_DIRECTIONAL_SELECTED_MATRIX_ACTIVE | noStrictnessIncrease=true
GPL5175 annotation: FULL_GPL5175_ANNOTATION_REQUIRED_BEFORE_TIER1_PASS | fullPass=false
Verified-local Cox package: VERIFIED_LOCAL_COX_RERUN_CANDIDATE_RELAXED_MODE |
label=SECONDARY_AXIS_VERIFIED_LOCAL_COX_RERUN_TARGET_NOT_TESTABLE_WITH_CURRENT_DATA
Reproducibility receipt:
sha256-835e4db3f313e90e23f56594e808185d3fef2ba457647423aaaf415be61f001f |
replay=REPLAY_PACKAGE_SPEC_READY_CURRENT_RUN_EMBEDDED_JSON_ONLY
Method claim: The app parses cancer hypotheses, maps them to expression-survival axes,
runs relaxed first-pass event-rate/Cox screens, ranks directional signals, and packages
reproducibility receipts while blocking official proof/kill claims before Tier-2.
Biology claim: This axis is interpreted only as a local directional result in the current
selected-matrix screen.
Evidence readiness gate: BLOCKED_BY_MISSING_EVIDENCE |
missing=official_or_declared_probe_platform_mapping,probe_to_gene_mapping_receipt,statistical_fragility_receipt
Evidence readiness receipts: platform/probe=PARTIAL_MAPPING_REVIEW_ONLY |
outcomeResolved=true | covariateResolved=true |
phenotypeComparator=SURVIVAL_OUTCOME_MODEL_NO_PHENOTYPE_COMPARATOR_REQUIRED_OR_NOT_DECLARED
| replicationComparator=REPLICATION_COMPARATOR_NOT_DECLARED |
confounder=CONFOUNDER_FIELDS_AVAILABLE_EXECUTION_NOT_CONFIRMED | mycnCoxExecuted=false |
fragility=NO_EXECUTED_STATISTICAL_MODEL_RECEIPT
Axis continuation: NO_AXIS_CONTINUATION_FROM_CURRENT_VERDICT | next=Keep current
evidence-readiness verdict; no axis continuation is opened.
Operational readiness: BLOCKED_FROM_PROMOTION_BY_READINESS_GAPS — Resolve first:
official_or_declared_probe_platform_mapping, probe_to_gene_mapping_receipt,

statistical_fragility_receipt_hr_ci_p_fdr_n_events

Final hypothesis state v3.0.6.0: NOT_MEASURABLE_PROBE_AND_CONTEXT_BLOCKED | reason=Bulk selected matrix cannot test NECTIN2-TIGIT ligand-receptor engagement; missing probes and single-cell/functional context block measurement.

Decisive adjudication v3.0.6.5:

verdict=NOT_TESTABLE_CURRENT_BULK_DATA_KILLS_BULK_CHECKPOINT_CLAIM | claimKilled=false | localSignalSurvives=false | notTestable=true | action=Do not run pseudo-Cox or proxy promotion; acquire the missing platform/single-cell/functional data first.

Investment verdict v3.0.6.6: verdict=KILL_CURRENT_BULK_CHECKPOINT_CLAIM | killedClaim=bulk_selected_matrix_can_test_NECTIN2_TIGIT_ligand_receptor_checkpoint_function | killCriteria=5 | investCriteria=0 | next=Do not invest in bulk NECTIN2/TIGIT claim. Only a single-cell/functional checkpoint dataset can reopen it.

FINAL INVESTMENT DECISION v3.0.6.7: DO_NOT_INVEST_CLAIM_KILLED | claim=claim killed as written for current data |

killedClaim=bulk_selected_matrix_can_test_NECTIN2_TIGIT_ligand_receptor_checkpoint_function | criteria=kill:5/invest:0 | next=Do not invest in bulk NECTIN2/TIGIT claim. Only a single-cell/functional checkpoint dataset can reopen it. | probabilityReported=false

PREPRINT READINESS v3.0.6.8: PREPRINT_READY_CLAIM_KILLED_AS_WRITTEN | title=Hypothesis Stress-Test & Investment Decision Report v3.0.6.8 | killedAsWritten=true |

residualDataFollowUp=false | wetLabAllowed=false | probabilityReported=false

Publication data fuel v3.0.6.9: route=NECTIN2_TIGIT_NOT_MEASURABLE |

state=NECTIN2_TIGIT_SINGLE_CELL_FUNCTIONAL_REOPEN_GATES_INJECTED_BULK_CLAIM_KILLED | syntheticData=false | fakeReplication=false | defense=NECTIN2/TIGIT bulk claim is killed because the relevant checkpoint biology is ligand-receptor and single-cell/functional, not bulk selected-matrix Cox.

Pediatric cross-tumor scope v3.0.6.9.6: current=NEUROBLASTOMA_ONLY_SINGLE_COHORT_SCOPE | observed=neuroblastoma | crossTumorPromotionAllowed=false | wetLabPromotionAllowed=false

Pediatric expansion targets: medulloblastoma,Wilms tumor / nephroblastoma,Ewing sarcoma

Pediatric scope next action: After neuroblastoma, run the same stress-test template on medulloblastoma, Wilms tumor/nephroblastoma, and Ewing sarcoma; keep each as KILLED / NULL-LIKE / SURVIVING / NOT_MEASURABLE with separate receipts before any cross-tumor claim.

Pediatric scope boundary: Neuroblastoma-only evidence remains single-cohort and disease-scoped. Cross-pediatric-tumor credibility requires independent human datasets from at least three additional pediatric tumor families before any generalization, Tier-2 promotion, or wet-lab language.

Preprint boundary: This report supports a methods/resource-allocation claim only: supplied computational hypotheses are triaged as killed-as-written, watchlist-only, or data-only follow-up. It does not prove or disprove biology, does not recommend treatment, and does not establish clinical actionability.

Allowed manuscript sentence: The supplied claim is killed as written for the current data; the result should redirect resources away from the claimed mechanism rather than imply biological impossibility.

Direction: UNKNOWN | caveat=Relaxed first-pass screen only. Shows trial event-rate, MYCN-stratified and in-app Cox-review numbers for debugging. Not proof, not official support/kill, not clinical, not publication-grade, and not Tier-2 computational evidence.

Missing gates: official NECTIN2/PVRL2 full GPL5175 mapping,official TIGIT/CD226/CD96 full GPL5175 mapping,single-cell ligand-receptor engagement not available,Tier-2 not reached,full GPL5175 mapping required before official support/kill,independent replication required before computational evidence language

Claim guard: BLOCKED_COMPUTATIONAL_CLAIM_REQUIRES_TIER2 |
tier=TIER0_SLICE_REPRODUCIBLE_REVIEW_ONLY | officialPromotionAllowed=false |
experimentalPromotionAllowed=false | experimentalPromotionForKilledClaimAllowed=false |
dataFollowUpAllowed=false | computationalReplicationAllowed=false | wetLabAllowed=false |
officialClaimAllowed=false

Tier-0 proves replayability, not source truth.

Tier-1 proves source extraction, not biological truth.

Tier-2 is required before any computational evidence claim.